# PHYSICS 2 FINAL CHEAT SHEET | $k=8.99\times10^9$ N·m²/C² • $\varepsilon_0=8.85\times10^{-12}$ C²/N·m² • $\mu_0=4\pi\times10^{-7}$ T·m/A • $e=1.602\times10^{-19}$ C • $m_e=9.11\times10^{-31}$ kg • $m_p=1.67\times10^{-27}$ kg • $c=3.00\times10^8$ m/s • $g=9.81$ m/s² • 1 eV=$1.602\times10^{-19}$ J | ▶ EXAM-FLAGGED ★ Cheat formula

## ★★★ FINAL EXAM — STARRED PROBLEM TYPES ★★★

⚠ "NEVER USE WEBERS" — keep magnetic flux as T·m², NOT Wb (1 Wb = 1 T·m²)

### ① Insulating Sphere — E inside/outside

▶ Uniform Q in sphere radius R. **Inside (r<R):** $E = kQr/R^3$
**Outside (r≥R):** $E = kQ/r^2$

**Worked:** E=760 N/C at r=R/4 (inside): $kQ(R/4)/R^3 = kQ/(4R^2) = 760 \rightarrow kQ/R^2 = 3040$.
At r=4R (outside): $E = kQ/(4R)^2 = kQ/(16R^2) = 3040/16 =$ **190 N/C**

*If asked at r=R: both formulas give $E=kQ/R^2$*

### ② Equivalent Resistance

▶ **Series:** $R_{eq} = R_1+R_2+R_3\ldots$
**Parallel:** $1/R_{eq} = 1/R_1+1/R_2+\ldots$
**Two parallel:** $R_{eq} = R_1R_2/(R_1+R_2)$

▶ **Method:** Reduce innermost groups first → simplify outward → $I_{total} = \varepsilon/R_{eq}$ → back-substitute V=IR for each branch.

▶ **Same R in parallel** (n of them): $R_{eq} = R/n$

### ③ RC Circuit (timing)

▶ **Charging:** $q(t) = Q_{max}[1-e^{-t/RC}]$, $Q_{max}=C\varepsilon$
$i(t) = (\varepsilon/R)\cdot e^{-t/RC}$
$V_C(t) = \varepsilon(1-e^{-t/RC})$; $V_R = \varepsilon\cdot e^{-t/RC}$

▶ **At t=0:** $V_C=0$, $V_R=\varepsilon$(MAX), $i=\varepsilon/R$(MAX), q=0
**At t=∞:** $V_C=\varepsilon$(MAX), $V_R=0$, i=0, $q=Q_{max}$
**τ = RC** (63% of final)

▶ **Discharging:** $q=Q_0e^{-t/RC}$; $i=(Q_0/RC)e^{-t/RC}$; $V=(Q_0/C)e^{-t/RC}$

*Correction: exponent is −t/RC (not −ε/RC)*

### ④ B from Moving Charge

▶ **$B = (\mu_0/4\pi)(qv\times\hat{r})/r^2$**
$|B| = (\mu_0/4\pi)\cdot(qv\sin\theta)/r^2$

▶ θ = angle between v and r. RHR for direction.

### ⑤ Long Straight Wire

▶ **$B = \mu_0I/(2\pi r)$** (outside, dist r from axis)

▶ **Direction:** RHR — thumb along I, fingers curl in B direction

▶ Inside (r<R): $B = \mu_0Ir/(2\pi R^2)$

### ⑥ Ampere's Law (Multiple Currents)

▶ **$\oint B\cdot dl = \mu_0\cdot I_{encl}$**

▶ **Sign rule:** Curl right hand along path direction. Thumb = + current direction.
★ **CCW path** → + out of page
★ **CW path** → + into page

▶ **$I_{encl}$** = sum with signs: $I_1(\pm) + I_2(\pm) + I_3(\pm)$

▶ **Then:** $B(2\pi r) = \mu_0I_{encl} \rightarrow B = \mu_0I_{encl}/(2\pi r)$

### ⑦ Toroid B-field

▶ **$B = \mu_0NI/(2\pi r)$** (inside, r = dist from CENTER axis)

▶ N = total turns. Outside: B = 0

⚠ **Flux $\Phi_B$ = BA in T·m², not Wb**

### ⑧ Faraday's Law of Induction

▶ **$\varepsilon = -d\Phi_B/dt$** $\Phi_B = BA\cos\theta$

▶ **If B(t) varies, θ const:** $\varepsilon = -A\cos\theta\cdot(dB/dt)$
**If θ(t) varies, B const:** $\varepsilon = BA\sin\theta\cdot(d\theta/dt)$
**Both vary:** $\varepsilon = -A[\cos\theta\cdot dB/dt - B\sin\theta\cdot d\theta/dt]$

▶ **Rotating coil:** $\theta=\omega t \rightarrow \varepsilon = NBA\omega\sin(\omega t)$; $\varepsilon_{max} = NBA\omega$

▶ **N loops:** multiply ε by N

### ⑨ Slide-Wire Generator

▶ **ε = vBL** (motional EMF)

▶ v = rod speed, B = field, L = rod length perp to v & B

▶ **Induced current:** $I = \varepsilon/R = vBL/R$
**Force to push:** $F = BIL = B^2L^2v/R$
**Power:** $P = I^2R = (vBL)^2/R$

### ⑩ RL Circuit

▶ **Energizing:** $i(t) = I_{max}[1 - e^{-Rt/L}]$; $I_{max}=\varepsilon/R$
**Decay:** $i(t) = I_0\cdot e^{-Rt/L}$

▶ **τ = L/R** | $V_L(t) = \varepsilon\cdot e^{-Rt/L}$ (energizing)

▶ **At t=0:** i=0 (inductor blocks). **t=∞:** i=ε/R

### ⑪ EM Waves — Full Set

▶ **$E(x,t) = E_m\sin(kx - \omega t)$**
**$B(x,t) = B_m\sin(kx - \omega t)$**

▶ **Amplitude ratio:** $E_m/B_m = c$
**Instant ratio:** $E(t)/B(t) = c$

▶ **Wave speed:** $c = 1/\sqrt{\mu_0\varepsilon_0} = 3\times10^8$ m/s
**Wave number:** $k = 2\pi/\lambda$ [rad/m]
**Angular freq:** $\omega = 2\pi/T = 2\pi f$ [rad/s]
**c = fλ = ω/k**

▶ **Poynting:** $S = (1/\mu_0)(\vec{E}\times\vec{B})$ [W/m²]
$|S| = EB/\mu_0 = E^2/(c\mu_0) = cB^2/\mu_0$
**Avg I:** $I = E_mB_m/(2\mu_0) = \frac{1}{2}c\varepsilon_0E_m^2$

▶ **Direction:** $\vec{E}\times\vec{B}$ = propagation (RHR)

## CH 21–24 CONDENSED (1 Q)

### Coulomb & E-fields

**$F = kq_1q_2/r^2$; $E = kQ/r^2$** (point)
**Inf line:** $E=\lambda/(2\pi\varepsilon_0r)=2k\lambda/r$
**Inf sheet:** $E=\sigma/(2\varepsilon_0)$; **Conductor:** $E=\sigma/\varepsilon_0$
**Ring axis:** $E=kqz/(z^2+a^2)^{3/2}$
★ Square ctr alt±q: $E=4\sqrt{2}\cdot kq/a^2$
★ Semicircle ½±Q: $E=4kQ/(\pi a^2)$
★ Zero-E pt (2 like q): $x=s/(1+\sqrt{q_2/q_1})$
★ Two ∥ sheets same sign: between=$|\sigma_A-\sigma_B|/(2\varepsilon_0)$; outside=$(\sigma_A+\sigma_B)/(2\varepsilon_0)$. Opp sign: swap.

### Dipole

$\vec{\tau}=\vec{p}\times\vec{E}$; $|\tau|=pE\sin\theta$
$U=-\vec{p}\cdot\vec{E}=-pE\cos\theta$; $F_{net}=0$ unif

### Gauss

**$\Phi=\oint E\cdot dA=Q_{encl}/\varepsilon_0$**
Inside unif sphere: $E=kQr/R^3=\rho r/(3\varepsilon_0)$
Inside unif cyl: $E=\rho r/(2\varepsilon_0)$
★ Cavity in sphere: $E=\rho\vec{h}/(3\varepsilon_0)$ UNIFORM

### Potential

$V=kQ/r$; $U=qV$; $\Delta V=-\int E\cdot dl$; $E=-dV/dx$
$W_{field}=q\Delta V=-\Delta U$; $\frac{1}{2}mv^2=q\Delta V$ (rest)
Plates: $E=V/d$. Line: $\Delta V=2k\lambda\ln(r_0/r)$

### Capacitors

$Q=CV$; $C=\varepsilon_0A/d$ (∥-plate); $\kappa\varepsilon_0A/d$ w/diel
Series: $1/C_{eq}=\Sigma1/C_i$; Par: $C_{eq}=\Sigma C_i$
**$U=\frac{1}{2}CV^2=Q^2/(2C)=\frac{1}{2}QV$**; $u=\frac{1}{2}\varepsilon_0E^2$
★ Discon (Q const): $U_{new}/U_{old}=d_{new}/d_{old}$
Connected (V const): $U_{new}/U_{old}=d_{old}/d_{new}$

## CH 25 — EMF & CIRCUITS

**$I=\varepsilon/(R+r)$**; $V_{term}=\varepsilon-Ir$ (discharge)
**V=IR; $P=IV=I^2R=V^2/R$**; $W=\varepsilon It$
★ Aiding batt: $I=(\varepsilon_1+\varepsilon_2)/\Sigma R$; Opp: $I=|\varepsilon_1-\varepsilon_2|/\Sigma R$ (dir = bigger ε wins)

### Current Density & Resistivity

**$I = nqv_dA$** (n=charge/vol, $v_d$=drift)
**$J = I/A = nqv_d$** [A/m²]
**$R = \rho L/A$**; ρ=resistivity [Ω·m]; σ=1/ρ
**$E = \rho J$** (microscopic Ohm's)
**Temp:** $\rho(T)=\rho_0[1+\alpha(T-T_0)]$

## CH 26 — KIRCHHOFF & RC

**Loop:** ΣV=0; **Junction:** $\Sigma I_{in}=\Sigma I_{out}$
**R-series:** $R_{eq}=\Sigma R_i$; **R-par:** $1/R_{eq}=\Sigma1/R_i$
$P_{dissipated}$ by R = $I^2R = V^2/R$

### ★ Find Time Algebra (RC/RL)

★ **Charging $V_C=\alpha\varepsilon$:** $t = -RC\cdot\ln(1-\alpha)$
α=½ → $t=RC\cdot\ln(2)\approx0.693RC$
★ **Decay to fraction f of $I_0$:** $t = -\tau\cdot\ln(f)$
(works RC and RL with τ=RC or L/R)
★ **RL energize to fraction f of $I_{max}$:** $t = -(L/R)\cdot\ln(1-f)$
★ **τ table:** 1τ→63%, 2τ→86%, 3τ→95%, 5τ→99.3%

### ★ Energy-Storage Cheat

| Element | Energy | Density |
|---|---|---|
| Capacitor | $\frac{1}{2}CV^2=Q^2/2C$ | $u=\frac{1}{2}\varepsilon_0E^2$ |
| Inductor | $\frac{1}{2}LI^2$ | $u=B^2/(2\mu_0)$ |
| EM Wave | — | $u=\varepsilon_0E^2=B^2/\mu_0$ |

*In EM wave: $u_E = u_B$ (equal split)*

### ★ Math Quick-Ref

**Derivatives:** $d/dt(e^{at})=ae^{at}$; $d/dt(\sin\omega t)=\omega\cos\omega t$; $d/dt(\cos\omega t)=-\omega\sin\omega t$

**Vectors:** $A\cdot B=AB\cos\theta$ (scalar); $A\times B=AB\sin\theta$ (vector ⊥ both, RHR)

**Geometry:** Circle area=$\pi r^2$; Sphere area=$4\pi r^2$, vol=$(4/3)\pi r^3$; Cylinder side=$2\pi rL$; Annulus=$\pi(R_2^2-R_1^2)$

**Quadratic:** $x = (-b\pm\sqrt{b^2-4ac})/(2a)$

**Logs:** $\ln(e^x)=x$; $\ln(ab)=\ln a+\ln b$; $\ln(a/b)=\ln a-\ln b$; $\ln(1)=0$; $\ln(2)\approx0.693$

**Trig:** $\sin^2+\cos^2=1$; $\sin(2\theta)=2\sin\theta\cos\theta$; $\cos(2\theta)=\cos^2\theta-\sin^2\theta$

**Common angles:** sin/cos at 0°=0/1; 30°=½/(√3/2); 45°=√2/2; 60°=(√3/2)/½; 90°=1/0

## CONSTANTS & CONVERSIONS

**$k = 1/(4\pi\varepsilon_0) = 8.99\times10^9$** N·m²/C²
**$\varepsilon_0 = 8.854\times10^{-12}$** C²/(N·m²)
**$\mu_0 = 4\pi\times10^{-7} = 1.257\times10^{-6}$** T·m/A
**$\mu_0/(4\pi) = 10^{-7}$** T·m/A
**$c = 1/\sqrt{\mu_0\varepsilon_0} = 2.998\times10^8$** m/s
**$e = 1.602\times10^{-19}$** C
**$m_e = 9.109\times10^{-31}$** kg = 0.511 MeV/c²
**$m_p = 1.673\times10^{-27}$** kg = 938 MeV/c²
**g = 9.81** m/s²
**1 eV = $1.602\times10^{-19}$ J**
**1 N/C = 1 V/m**
**1 Wb = 1 T·m²** ⚠ DON'T USE Wb
**1 H = 1 V·s/A = 1 T·m²/A**
**1 Hz = 1/s**; 1 rad ≈ 57.3°
**1 G = $10^{-4}$ T**; 1 mT = $10^{-3}$ T

## UNITS REFERENCE

| Quantity | Sym | Unit |
|---|---|---|
| Charge | q,Q | C = A·s |
| E-field | E | N/C = V/m |
| Potential | V | V = J/C |
| Energy | U | J = C·V |
| Capacitance | C | F = C/V |
| Current | I | A = C/s |
| Curr density | J | A/m² |
| Resistance | R | Ω = V/A |
| Resistivity | ρ | Ω·m |
| Power | P | W = J/s |
| EMF | ε | V |
| B-field | B | T = N/(A·m) |
| **Mag flux** ⚠ | $\Phi_B$ | **T·m²** (NOT Wb) |
| Inductance | L | H = V·s/A |
| Reactance | $X_L, X_C$ | Ω |
| Impedance | Z | Ω |
| Mag moment | μ | A·m² = J/T |
| Frequency | f | Hz = 1/s |
| Angular freq | ω | rad/s |
| Wavelength | λ | m |
| Wave number | k | rad/m |
| Period | T | s |
| Energy dens | u | J/m³ |
| Intensity | I,S | W/m² |
| Rad pressure | $P_{rad}$ | Pa = N/m² |
| Linear chg | λ | C/m |
| Surface chg | σ | C/m² |
| Volume chg | ρ | C/m³ |
| Permittivity | $\varepsilon_0$ | F/m |
| Permeability | $\mu_0$ | H/m |
| Dipole moment | p | C·m |
| Drift vel | $v_d$ | m/s |
| Conductivity | σ | S/m |

| SI | p | n | μ | m | c | k | M | G |
|---|---|---|---|---|---|---|---|---|
| 10^ | −12 | −9 | −6 | −3 | −2 | 3 | 6 | 9 |

## ★ WORKED PRACTICE EXAMPLES & DECISION TREE

### WE-1 ▸ RC Charging

**R=10 kΩ, C=100 μF, ε=12 V. Find t for $V_C$=8 V.**
τ = RC = **1.0 s**; $8 = 12(1-e^{-t/\tau}) \rightarrow e^{-t/\tau} = 1/3$
$t = -\tau\ln(1/3) =$ **1.10 s**

### WE-2 ▸ Long Wire B

**I=5 A, r=2 cm.**
$B = \mu_0I/(2\pi r) = (10^{-7}\cdot10)/(0.02) =$ **$5\times10^{-5}$ T**
Sanity: ≈ Earth's B (50 μT) ✓

### WE-3 ▸ Mass Spec

**V=1000 V, B=0.5 T, R=0.05 m.**
$m/q = B^2R^2/(2V) = (0.25)(0.0025)/2000 =$ **$3.13\times10^{-7}$ kg/C**
Singly-ionized: m ≈ **$5\times10^{-26}$ kg** (~30 amu)

### WE-4 ▸ Slide-Wire

**B=0.4 T, L=0.3 m, v=2 m/s, R=5 Ω.**
ε = BLv = **0.24 V**; I = **48 mA**
$F_{push} = B^2L^2v/R =$ **5.76 mN**; P = εI = **11.5 mW**

### WE-5 ▸ Rotating Coil

**N=50, A=0.01 m², B=0.2 T, ω=377 rad/s.**
$\varepsilon_{max} = NBA\omega =$ **37.7 V (peak)**
$V_{rms} = 37.7/\sqrt{2} =$ **26.7 V**

### WE-6 ▸ Inductor Energize

**ε=24 V, R=12 Ω, L=0.6 H. Find i at t=0.05 s.**
τ = L/R = 0.05 s (one tau!); $I_{max}$ = 2 A
$i(\tau) = I_{max}(1-e^{-1}) =$ **1.26 A**

### WE-7 ▸ LRC Resonance + EM Wave

**L=0.1 H, C=10 μF:** $\omega_0 = 1/\sqrt{LC} =$ **1000 rad/s**; $f_0 =$ **159 Hz**
**EM wave $E_m$=100 V/m:** $B_m = E/c =$ **$3.33\times10^{-7}$ T**; $I = \frac{1}{2}c\varepsilon_0E_m^2 =$ **13.3 W/m²**

### ★ COMMON PRODUCTS

**$\mu_0/(4\pi) = 10^{-7}$** T·m/A | **$Z_0 = 1/(c\varepsilon_0) \approx 377$ Ω**
**$1/\sqrt{2} \approx 0.707$; $e^{-1} \approx 0.368$; $\pi^2 \approx 9.87$**
**ln(2) ≈ 0.693; ln(10) ≈ 2.303; ln(3) ≈ 1.099**

### ★ DECISION TREE — Pick Formula Fast

- **Charge in B-field, find r** → r = mv/(qB)
- **Two parallel wires** → $F/L = \mu_0I_1I_2/(2\pi d)$
- **Loop moves through B** → ε = BLv
- **B(t) changing, loop fixed** → ε = −A(dB/dt)
- **Loop rotating in const B** → ε = NBAω sin(ωt)
- **Inductor at t=0:** i=0 (open) | **at t=∞:** i=ε/R
- **Capacitor at t=0:** V=0 (wire) | **at t=∞:** i=0
- **AC find Z** → $\sqrt{R^2+(X_L-X_C)^2}$
- **AC resonance ω** → $1/\sqrt{LC}$
- **EM wave: E given, find B** → B = E/c
- **EM wave intensity** → $\frac{1}{2}c\varepsilon_0E_m^2$
- **Cap connected, d↑** → V const, U↓
- **Cap disconnected, d↑** → Q const, U↑
- **Wire R from material** → R = ρL/A
- **Power dissipated** → $I^2R = V^2/R$